Moon-ki Choi

Postdoctoral Researcher

Materials Research Laboratory, *University of Illinois Urbana-Champaign*

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Education

2018–2023 **Ph.D. in Aerospace Engineering and Mechanics**
University of Minnesota Twin Cities, MN, USA
- Advisor: Prof. Ellad B. Tadmor
- Thesis: Atomistic and continuum simulations of 2D materials with environmental effects

2016–2018 **M.S. in Mechanical Engineering**
Sungkyunkwan University, South Korea
- Advisor: Prof. Moon Ki Kim
- Thesis: Atomic simulation of membrane and its bio-nano organelles

2009–2016 **B.S. in Mechanical Engineering**
Sungkyunkwan University, South Korea

Research Interests

Multiscale Modeling *Continuum mechanics; Molecular dynamics; Quantum mechanics*
Integrated simulation frameworks that connect quantum, atomistic, and continuum scales for predicting mechanical and electronic properties of materials.

Quantum Materials *2D materials; Topological insulator; Strain and defect engineering*
Atomic-scale defects, large-scale strain fields, and electronic structure of quantum materials.

Machine Learning for Materials *Equivariant graph neural networks; Gaussian regression*
Physics-informed machine learning models to accelerate multiscale simulations cosidering uncertainty.

Research Experience

2023–present **Postdoctoral Researcher**
University of Illinois Urbana–Champaign, IL, USA
- Advisor: Prof. Harley T. Johnson

2019–2023 **Research Assistant**
University of Minnesota Twin Cities, MN, USA

2018–2019 **Teaching Assistant**
University of Minnesota Twin Cities, MN, USA

Summer 2017	**Visiting Student**
	University of Cincinnati, OH, USA
2016–2018	**Research Assistant**
	Sungkyunkwan University, South Korea
	Material Strength & Computational Bioengineering Laboratory
Fall 2015	**Undergraduate Researcher**
	Sungkyunkwan University, South Korea
	Material Strength & Computational Bioengineering Laboratory

Skills

Computational Language	Python, C++, Fortran90, MATLAB
Quantum Mechanics Simulation	Quantum Espresso, OpenMX, Wannier90, Pythtb, ASE
Molecular Dynamics Simulation	LAMMPS, GROMACS, ASE
Continuum Model Simulation	Abaqus, FEniCS
Miscellaneous	Linux, Git, LaTeX, OpenMP

Publications

# indicates co-first authorship.

Under Review/In Revision

[1] S. Frisone, **M. Choi**, J. Hung, J. Essman, Z. You, S. Sinha, A. Birge, H. T. Johnson, M. L. Lee, C. Uher, C. Kurdak, and R. S. Goldman, "**Mechanisms for stress relaxation in BiSb single crystals**", *Submitted (Journal of Applied Physics).*

[2] A. Liu, A. Gil, **M. Choi**, B. A. Hanedar, Z. You, S. Sinha, T. Hakioglu, H. T. Johnson, K. Sun, R. Clarke, C. Uher, C. Kurdak, and R. S. Goldman, "**Probing topological surface states and conduction via extended defects in $(Bi_{1-x}Sb_x)_2Te_3$ films**", *Under review (Applied Physics Letters Materials).*

[3] **M. Choi** and E. B. Tadmor, "**Statistical mechanics of a 2D material in a gas reservoir**", *In revision (Journal of Chemical Theory and Computation).*

[4] **M. Choi**#, P. Jisuk#, and K. Junpyo, "**Atomically interlocked covalent organic frameworks as metamaterials: a molecular dynamics study of COF-500**", *In revision (Journal of the American Chemical Society).*

[5] S. Sinha, Z. You, S. Smolenski, E. Downey, A. Liu, S. Frisone, A. Birge, **M. Choi**, A. A. Shawon, E. Chandler, C. Lin, M. Hashimoto, D. Lu, M. L. Lee, H. T. Johnson, N. H. Jo, C. Uher, R. S. Goldman, and C. Kurdak, "**Emerging conduction pathways in semiconducting bismuth-antimony alloys**", *Under review (Journal of Physics: Materials).*

Preprint

[1] **M. Choi**, D. Palmer, and H. T. Johnson, "**Graph neural network for unified electronic and interatomic potentials: strain-tunable electronic structures in 2D materials**", *arXiv preprint*, https://doi.org/10.48550/arXiv.2510.16605 (2025).

[2] **M. Choi**, D. Palmer, and H. T. Johnson, "**Interatomic potential development for topological insulator $Bi_{1-x}Sb_x$ and its dislocation by force-following active learning**", *arXiv preprint*, https://doi.org/10.48550/arXiv.2510.18217 (2025).

Published/Accepted

[1] M. Kim[#], S. Jung[#], S. Kim[#], **M. Choi[#]**, J.-H. Hong, K. Kim, C. Park, K. J. Yu, G. D. Cha, S.-H. Sunwoo, Q. Liu, D.-H. Kim, and T.-W. Kim, "**2D silver nanosheet assembly for an isotropic, stretchable, and highly conductive nanomembrane**", *Advanced Materials*, e16002 (2025).

[2] M. T. Ahmed, **M. Choi**, H. T. Johnson, and N. C. Admal, "**Quantifying superlubricity of bilayer graphene from the mobility of interface dislocations**", *ACS Applied Materials & Interfaces* 17, 30197–30211 (2025).

[3] S. Edward, **M. Choi**, and H. T. Johnson, "**Application of time series analysis on kinetic monte carlo simulations of hyperthermal oxidation of graphite**", *The Journal of Physical Chemistry C* 129, 1692–1701 (2025).

[4] **M. Choi** and H. T. Johnson, "**Mechanics of out-of-plane screw dislocation in a 2D material**", *Extreme Mechanics Letters* 77, 102320 (2025).

[5] S. W. Park, **M. Choi**, B. H. Lee, S. Seo, W. K. Kim, and M. K. Kim, "**An accelerated molecular dynamics study for investigating protein pathways using the bond-boost hyperdynamics method**", *Protein Science* 34, e70073 (2025).

[6] J. R. Gissinger, I. Nikiforov, Y. Afshar, B. Waters, **M. Choi**, D. S. Karls, A. Stukowski, W. Im, H. Heinz, A. Kohlmeyer, and E. B. Tadmor, "**Type label framework for bonded force fields in lammps**", *The Journal of Physical Chemistry B* 128, 3282–3297 (2024).

[7] **M. Choi**, S. H. Sung, R. Hovden, and E. B. Tadmor, "**Elastic plate basis for the deformation and electron diffraction of twisted bilayer graphene on a substrate**", *Physical Review B* 110, 024116 (2024).

[8] **M. Choi[#]**, M. Pasetto[#], Z. Shen[#], E. B. Tadmor, and D. Kamensky, "**Atomistically-informed continuum modeling and isogeometric analysis of 2D materials over holey substrates**", *Journal of the Mechanics and Physics of Solids* 170, 105100 (2023).

[9] Y. Zhang[#], **M. Choi[#]**, G. Haugstad, E. B. Tadmor, and D. J. Flannigan, "**Holey substrate-directed strain patterning in bilayer MoS_2**", *ACS Nano* 15, 20253–20260 (2021).

[10] H. Kim, B. H. Lee, **M. Choi**, S. Seo, and M. K. Kim, "**Effects of aquaporin-lipid molar ratio on the permeability of an aquaporin z-phospholipid membrane system**", *PLOS One* 15, e0237789 (2020).

[11] T. Kim, **M. Choi**, H. S. Ahn, J. Rho, H. M. Jeong, and K. Kim, "**Fabrication and characterization of zeolitic imidazolate framework-embedded cellulose acetate membranes for osmotically driven membrane process**", *Scientific Reports* 9, 5779 (2019).

[12] B. H. Lee, S. Jo, **M. Choi**, M. H. Kim, J. B. Choi, and M. K. Kim, "**Normal mode analysis of zika virus**", *Computational Biology and Chemistry* 72, 53–61 (2018).

[13] C. S. Lee[#], **M. Choi**[#], Y. Y. Hwang, H. Kim, M. K. Kim, and Y. J. Lee, "**Facilitated water transport through graphene oxide membranes functionalized with aquaporin-mimicking peptides**", *Advanced Materials* 30, 1705944 (2018).

[14] **M. Choi**, H. Kim, B. H. Lee, T. Kim, J. Rho, M. K. Kim, and K. Kim, "**Understanding carbon nanotube channel formation in the lipid membrane**", *Nanotechnology* 29, 115702 (2018).

[15] B. H. Lee, S. Seo, M. H. Kim, Y. Kim, S. Jo, **M. Choi**, H. Lee, J. B. Choi, and M. K. Kim, "**Normal mode-guided transition pathway generation in proteins**", *PLOS One* 12, e0185658 (2017).

[16] **M. Choi**, S. Jo, B. H. Lee, M. H. Kim, J. B. Choi, K. Kim, and M. K. Kim, "**Dynamic characteristics of a flagellar motor protein analyzed using an elastic network model**", *Journal of Molecular Graphics and Modelling* 78, 81–87 (2017).

Research Funding Experience

National high-performance computing resource program National Science Foundation (NSF)	***Role: PI***	1,900,000 Service Units (≈ **1,900,000 CPU hours**) awarded	2024–2028
National high-performance computing resource program National Science Foundation (NSF)	***Role: PI***	4,200,000 Service Units (≈ **4,200,000 CPU hours**) awarded	2024–2026
Early Career Research Program National Research Funding, South Korea	***Role: Co-PI***	Proposal preparation and submission	2025
Scholarship and Teaching for Engineering Postdocs Program University of Illinois Urbana–Champaign	***Role: Trainee***	Training in research proposal writing and grant development	2024

Awards and Honors

Best Poster Prize (sponsored by *Journal of Materials Science: Material Theory*, $200)	*Nanomaterials 2025, U.S. National Congress on Computational Mechanics (USACM)*	2025
Travel Award	*Nanomaterials 2025, U.S. National Congress on Computational Mechanics (USACM)*	2025
Doctoral Dissertation Fellowship ($25,000 + tuition)	*University of Minnesota Twin cities*	2023
Outstanding Poster Award	*Korea MultiScale Mechanics Society*	2017
Outstanding Poster Award	*Korean BioChip Society*	2017
Excellence Award for Oral Presentation	*Sungkyunkwan University*	2016